DREAM MACHINES OF THE FUTURE: Computers Based On Cutting-Edge Technology

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"Computers in the future may weigh no more than 1.4 tonnes."
—Popular Mechanics, 1949

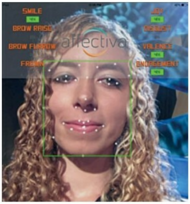
"There is no reason anyone would want a computer in their home."
—Ken Olson, president, chairman and founder of Digital Equipment Corp.

The two quotes show us how unpredictable the future is and that we probably have no idea what is in store for us in terms of the size, power, capabilities, etc of computers in, say, 50 years from now. However, let us try and imagine how computer technology is likely to shape up in about 15 years. Let us try and be as creative as possible to paint a possible future scenario, perhaps based on our wish lists, needs and fantasies.

Computers with an emotion chip could find their way into everyday things used by us. For example, your mirror could sense that you are in a bad mood, communicate this to your best friend and request her to give you a call to comfort you.

Fig. 1: A screen shot from an emotion diagnostic software at work (Image courtesy: http://hothardware.com)

The year is 2030. Your computer has an emotion chip, which, with the help of various sensors like optics, heat and heartbeat rate, is able to gauge your mood. For example, your computer knows when you have a frown on your face; tone of your voice is irritated and your heart is beating faster than normal. This chip then triggers some useful actions that could lower the temperature of the room, and delivers a voice message in the kitchen that a glass of cold water should be sent to you quickly, and so on.



Your personal network

In the future, it is expected that a strong personal network will evolve. Mostly, the personal network will comprise one or more computing devices in communication with one another with the help of various permanent and transient communication links. One or more devices would be very high-end laptops—the basic portal through which you, as a user, will access the network. Once disconnected from the personal network, this portal will work just like a standalone device.

The speed of wireless Internet would be so high that actual computing and data processing will not be on your personal device, but on a far off, more powerful machine connected via the Cloud. Such a technology will enable you to process data at lighting speeds. For example, you could be researching the way proteins take certain shapes, called folds, and how that relates to what proteins do.

This problem that typically takes months at present will be completed in just a few seconds. Problems could range from breaking codes to searching data sets and even extend to the creation of new materials.

Now, close your eyes, step into the time machine and take yourself to the 2030s. Your personal network has notified the navigation computer in your car about your departure time and destination, today morning. Just as you are backing out of the driveway, the navigation computer is informed of a traffic jam, and it informs you of an alternate route, estimates your new arrival time, notifies your office staff of your situation and reschedules your early meetings.

You are now driving to work. How-